Large Scale QKD Sim

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Assumptions

1) QKD not happening. QKD Keys are simulated
2) HTTPS for REST. Paper uses HTTP

Components

1) Node - User interaction, QKD module, KMS, Controller etc
2) KMS/KME - Key mangement Sys/Entity.
   - Provides instructions to manage entire key vault
   - The big brain behind all the ops
3) QBuffer - A Buffer store that rcv ~~data~~ keys after QKD + post process
   - QKD Module → Post Process → KMS → QBuffer → KMS → SBuffer
   - Primary task: Ensure that sys never runs out of QKD key materia
4) SBuffer - Key pool populated from QBuffer
   - Types:
     - Local: Created in enc & dec to store keys
     - Relay: Has Key material when it needs to be shared to an indirect nd
     - Stream: Has key material that are being used RN.
5) SAE - Secure Application Layer - the app that uses key material

KMS Synch

1) KMS/KME of 2 nodes are synchronised with each other.
2) The in-sync parts have corresponding copies of QKD gen key material.
3) KMS are directional. $SBuffer_A^{Local(Enc)} \equiv SBuffer_B^{Local(End)}$

ETSI QKD014

1) REST Based comm b/w SAE & KMS

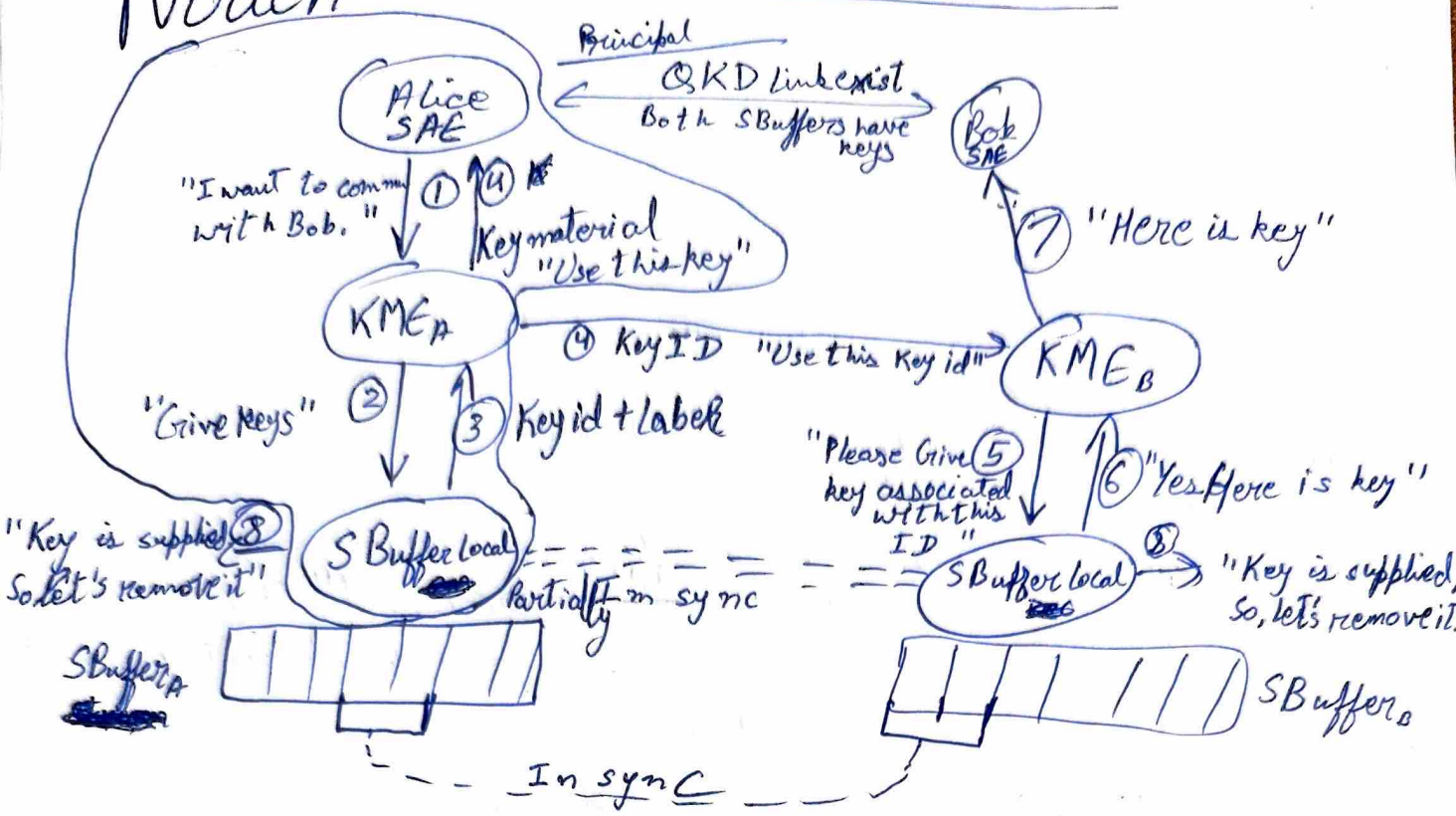
ETSI QKD 004

1) Session key streamer. Based on QoS.

Quantum Walk
TF QKD
Distillationless

Node A

Alice Bob Direct QKD Link



In direct Alice-Bob

